

Nion Rahaman Akash

B.Sc. in Mathematics | Postgraduate Diploma in Data Science

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Dhaka, Bangladesh

RESEARCH INTERESTS

Data Science | Machine Learning – Trustworthy and Reliable | Health Care | Natural Language Processing for Low-Resource Languages

EDUCATION

Postgraduate Diploma (PGD) in Data Science

United International University (UIU), Bangladesh • 2024 • CGPA 3.44/4.00

- Key areas: Python, Machine Learning, Deep Learning, SQL, Data Analysis, Data Visualization

B.Sc. in Mathematics

Dept. of Mathematics, University of Rajshahi, Bangladesh • 2016 – 2022 • CGPA 2.96/4.00

- Major theory: Statistics, Operations Research, Real Analysis, Linear Algebra, Numerical Analysis, Differential Equations, Discrete Mathematics, Theory of Matrices, Computer Programming (Fortran 90, C)

MAJOR PROJECTS

- **Diabetes Risk Prediction** — Logistic Regression, Random Forest, and a Keras/TensorFlow neural network for diabetes risk classification; 75% accuracy. Data preprocessing, feature engineering, and model evaluation in Python.
- **Car & Laptop Price Prediction** — Regression pipelines (Linear, Ridge, Random Forest, Gradient Boosting) with full data-cleaning, cross-validated model comparison, and a fixed target-leakage bug.
- **Breast Cancer Classification** — Support Vector Machines (SVM) for tumor classification.
- **Digit Recognition** — Random Forest and K-Nearest Neighbors (KNN) on handwritten digit data.
- **Spam Detection & Sentiment Analysis** — NLP (Bag-of-Words, Naive Bayes) for text classification.
- **Customer Segmentation** — K-Means Clustering and RFM Analysis for customer behavior grouping.

Full code, write-ups, and results for these projects: github.com/Akash-Nion

CERTIFICATIONS

- R for Applied Statistics and Data Science — ISRT, University of Dhaka (36-hour intensive)
- Python for Data Science, AI & Development — Coursera

TECHNICAL SKILLS

- **Programming:** Python, R, C, Fortran, MATLAB
- **Machine Learning:** Regression, Classification, Clustering, Model Evaluation
- **Data Science Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Databases & Query Languages:** SQL, Neo4j, Cypher
- **Data Preparation:** Feature scaling, encoding techniques, outlier detection (IQR, Z-score)

EXTRACURRICULAR & LEADERSHIP

Volunteer, Transparency International Bangladesh (TIB)

Feb 2021 – Jan 2023

- Conducted primary data collection across rural and urban households for evidence-based social accountability research
- Promoted institutional transparency and ethical governance in public sectors (Health, Education) through anti-corruption initiatives
- Facilitated information dissemination campaigns translating governance protocols into accessible insights

Active Member, University of Rajshahi English Speaking Club

- Facilitated structured communication workshops, enhancing the public speaking and technical articulation skills of diverse peer groups
- Led collaborative learning sessions, fostering an environment of constructive feedback and continuous improvement in academic fluency

STANDARDIZED TEST

- **IELTS** (28 Jan 2026) — Overall 6.5 (Listening 7.5, Reading 6.5, Writing 6.0, Speaking 6.0)